

Project Report: Extracting Entities from Recipe Data using CRF

1. Problem Statement

The goal of this assignment was to build a Named Entity Recognition (NER) model to structure unstructured recipe text. A raw string like *"2 cups chopped onions"* needs to be broken down into specific entities: **Quantity** (2), **Unit** (cups), and **Ingredient** (onions). This structure is critical for applications like nutrition analysis apps or smart grocery lists.

2. Methodology

I used **Conditional Random Fields (CRF)**, a sequence modeling technique that is ideal for this task because it considers the context of a word (what came before and after it) rather than looking at words in isolation.

Key Steps:

- **Data Cleaning:** I identified and removed rows where the input text length didn't match the label length to prevent training errors.
- **Feature Engineering:** This was the most critical part. I created a custom `word2features` function that extracted:
 - **Linguistic Features:** POS tags, lemmas, and word shapes using **spaCy**.
 - **Domain Knowledge:** I created custom lists for known units (e.g., *tbsp*, *oz*) and regex patterns to catch fractions (e.g., *1/2*) and decimals.
 - **Context:** Features for the "previous word" and "next word" to help the model learn patterns like [Number] -> [Unit] -> [Ingredient].
- **Handling Imbalance:** Since "Ingredient" is a very common label, I calculated inverse frequency weights to penalize it and force the model to pay more attention to the rarer "Unit" and "Quantity" classes.

3. Model Results

The model was trained using the L-BFGS algorithm and evaluated on a 30% validation split.

- **Accuracy:** The model achieved high overall accuracy (>95%).

- **Confusion Matrix Analysis:** The matrix showed that the model separates classes well. The diagonal line was strong, indicating high True Positives.
- **Weighted F1-Score:** The usage of regex patterns significantly boosted the recall for the Quantity class, which is usually hard to detect due to fractions.

4. Error Analysis & Insights

By analyzing the specific errors in the validation set, I found:

- **Ambiguity is tricky:** Words like "clove" caused confusion because they can be a unit (clove of garlic) or an ingredient (ground cloves).
- **Formatting matters:** The model struggled when spaces were missing (e.g., "1cup"), treating it as a single unknown word.
- **Context:** The model is very dependent on the standard recipe structure. If a sentence was written in an unusual order, accuracy dropped.

5. Conclusion

The CRF model proved to be a robust solution for recipe extraction. While a simple rule-based system would fail on fractions or variations, the CRF learned to generalize well. Future improvements could include using a Bi-LSTM deep learning model to capture longer sentence dependencies.